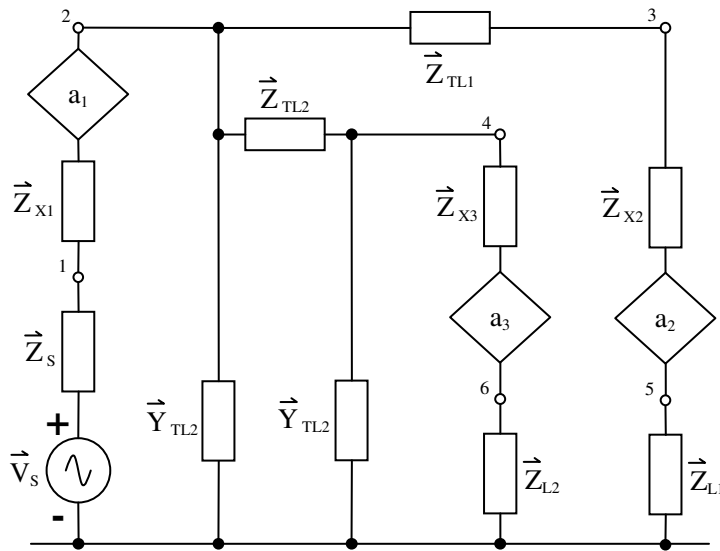


Problem #1



Problem #2

Step 4: Circuit Analysis

$$\vec{Z}_{eq1} = \vec{Z}_{TL} + (\vec{Z}_{L1}^{-1} + \vec{Z}_{L2}^{-1})^{-1} = 2.9177 \angle 11.061^\circ \Omega$$

$$\vec{I}_G = \vec{E}_G / (\vec{Z}_G + \vec{Z}_{eq1}) = 48.1654 \angle -11.440^\circ A$$

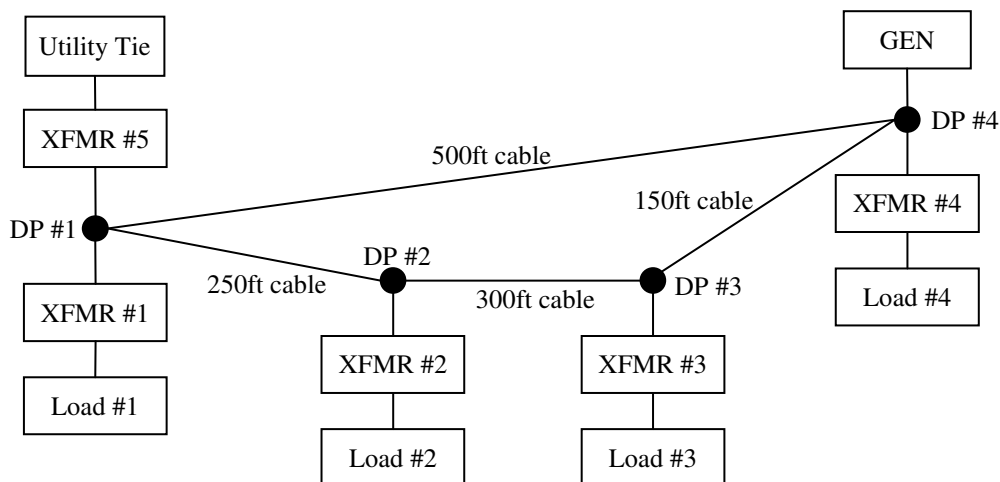
$$\vec{V}_1 = \vec{Z}_{eq1} \vec{I}_G = 140.5335 \angle -0.379^\circ V$$

$$\vec{S}_G = \vec{V}_1 \cdot \vec{I}_G^* = 6768.8525 \angle 11.061^\circ VA$$

Step 5: Final Results

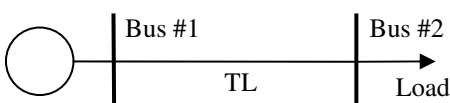
$$P_{G3\phi} = 3 \cdot \text{Re}\{\vec{S}_G\} = 19.9293 \text{ KW}, \quad Q_{G3\phi} = 3 \cdot \text{Im}\{\vec{S}_G\} = 3.9860 \text{ KVAR}$$

Problem #3

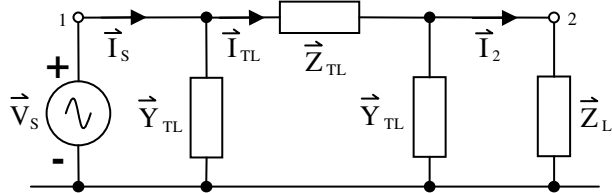


Problem #4

Step 1 – Visualize



Step 2 – Equivalent Circuit



Step 3 – Circuit Parameters

$$\vec{V}_s = 487 \angle 0^\circ \text{ V}, \quad P_s = 1212 \text{ W}, \quad Q_s = 2080 \text{ W}$$

$$\vec{Z}_{TL} = 2(400 \text{ ft})(0.0013 \angle 25^\circ \Omega / \text{ft}) = 1.04 \angle 25^\circ \Omega, \quad \vec{Y}_{TL} = 0.5(400 \text{ ft})(0.00001 \angle 87^\circ \text{ S} / \text{ft}) = 0.002 \angle 87^\circ \text{ S}$$

Step 4: Circuit Analysis

$$\vec{S}_s = P_s + jQ_s = 2407.3521 \angle 59.771^\circ \text{ VA}, \quad \vec{I}_s = (\vec{S}_s / \vec{V}_s)^* = 4.943228 \angle -59.771^\circ \text{ A}$$

$$\vec{I}_{TL} = \vec{I}_s - \vec{Y}_{TL} \vec{V}_s = 5.7827 \angle -65.067^\circ \text{ A}$$

$$\vec{V}_2 = \vec{V}_s - \vec{Z}_{TL} \vec{I}_{TL} = 482.4131 \angle 0.460^\circ \text{ V}, \quad \vec{I}_2 = \vec{I}_{TL} - \vec{Y}_{TL} \vec{V}_2 = 6.6536 \angle -68.903^\circ \text{ A}$$

$$\vec{S}_2 = \vec{V}_2 \cdot \vec{I}_2^* = 3209.7716 \angle 69.363^\circ \text{ VA}$$

Step 5: Final Results

$$\eta_{TL} = \left[\frac{\text{Re}\{\vec{S}_2\}}{P_s} \right] \times 100 = \left[\frac{1131.2973 \text{ W}}{1212 \text{ W}} \right] \times 100 = 93.341\%$$

Problem #5

$$S_{rated} = 500 \text{ KVA}, \quad V_{rated} = 480 \text{ V}, \quad \vec{Z}_G = 0.125 \angle 85^\circ \Omega$$

$$(base\Omega) = \frac{(rated \text{ KV})^2}{(rated \text{ MVA})} = \frac{(0.480)^2}{(0.500)} = 0.4608$$

$$Z_G \text{ in } \% = \left[\frac{(Z_G \text{ in } \Omega)}{(base\Omega)} \right] \cdot 100 = 27.0235\%, \quad X / R_G = \tan \theta_G = \tan 85^\circ = 11.4301$$

Problem #6

$$S_{3\phi rated} = 750 \text{ KVA}, \quad V_{LL1 rated} = 13.8 \text{ KV}, \quad \vec{Z}_x = 10.261 + j28.191 = 30.0003 \angle 69.999^\circ \Omega$$

$$(base\Omega) = \frac{(rated \text{ primary side KV})^2}{(rated \text{ MVA})} = \frac{(13.8)^2}{(0.750)} = 253.92$$

$$Z_x \text{ in } \% = \left[\frac{(Z_x \text{ in } \Omega)}{(base\Omega)} \right] \cdot 100 = 11.8149\%, \quad X / R_x = 28.191 / 10.261 = 2.7474$$